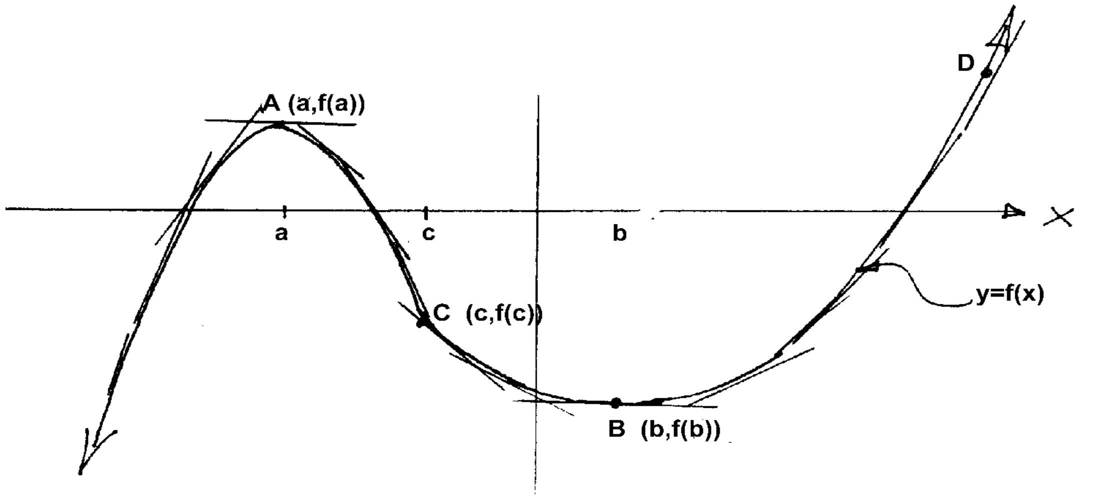
Maximum and Minimum Values

Knowledge of the slope of the tangent line to a curve (function) allows us to pinpoint locations of local maxima or minima. Also known as local extrema.



Point A (on the graph above) is termed a **relative or local maximum** point because in A's vicinity there are no points higher. D is higher but it is not in the vicinity of A. Similarly B is known as a **relative or local minimum** point, because in B's vicinity there is no point lower.

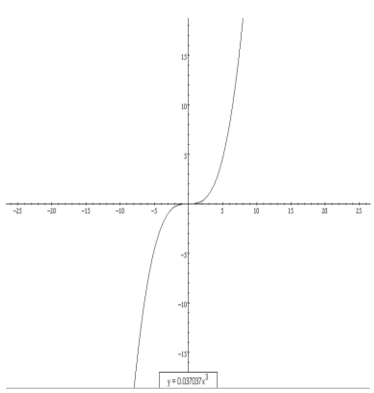
Critical Number: A number c in the domain of f is a critical number of f if either $f'(c)=0$ or $f'(c)$ does not exist. The point $(c, f(c))$ is called a **critical point**.

Fermat's Theorem: If f has a local maximum or minimum value at $x=c$, then either $f'(c)=0$ **or** $f'(c)$ does not exist.

<u>1st Derivative Test</u>

For a local maximum to occur, $f'(x)$ must change from ___________ to ______________ .

For a local minimum to occur, $f'(x)$ must change from ___________ to ______________ .



Note: Note the sign of $f'(x)$ must change for a local extrema.

Example: Consider $f(x)=x^3$. $(f'(x)=3x^2)$

When $x=0$, $f'(x)=$_____ .

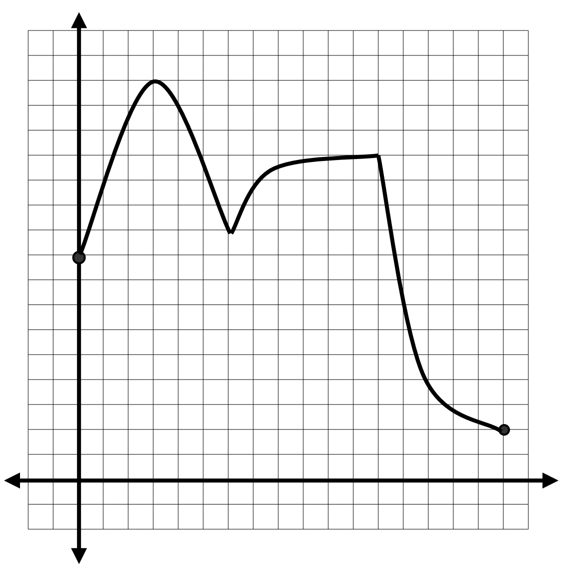
Yet (0,0) is obviously not a relative maximum nor minimum point. It may at best be described as a *"flat spot"*. Mathematically, it is called a "point of inflection" (POI).

Maximum and Minimum Values

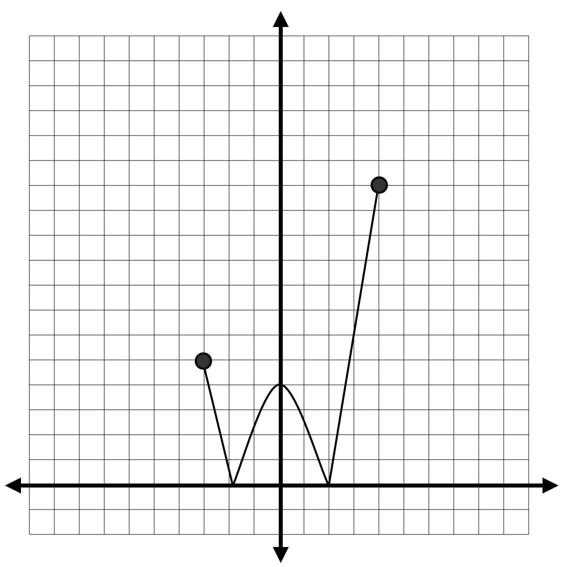
Example 1: Determine the coordinates of any local extrema for the function $f(x) = x^4 - x^3$.

To find absolute maximum or minimum points, you need to also test the endpoints of the domain.

Example 2: Determine the local and absolute extrema for each function:



$y = f(x)$ for $x \in [0,12]$



$y = \left|x^2 - 4\right|$ for $x \in [-3,4]$

Example 3: Determine the absolute maximum and absolute minimum values of the function $f(x) = 2x^4 - 4x^2 + 4$ for $x \in [-2, 0.5]$.

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Additional Questions

1. Determine the local max./min. of $f\left(x\right)=3x^5-5x^3$.

2. Given $f\left(x\right)=ax^3+bx^2+cx+d$, determine the values of a, b, c and d such that there is a local maximum at $(-2, 13)$ and a local minimum at $(0, 5)$.